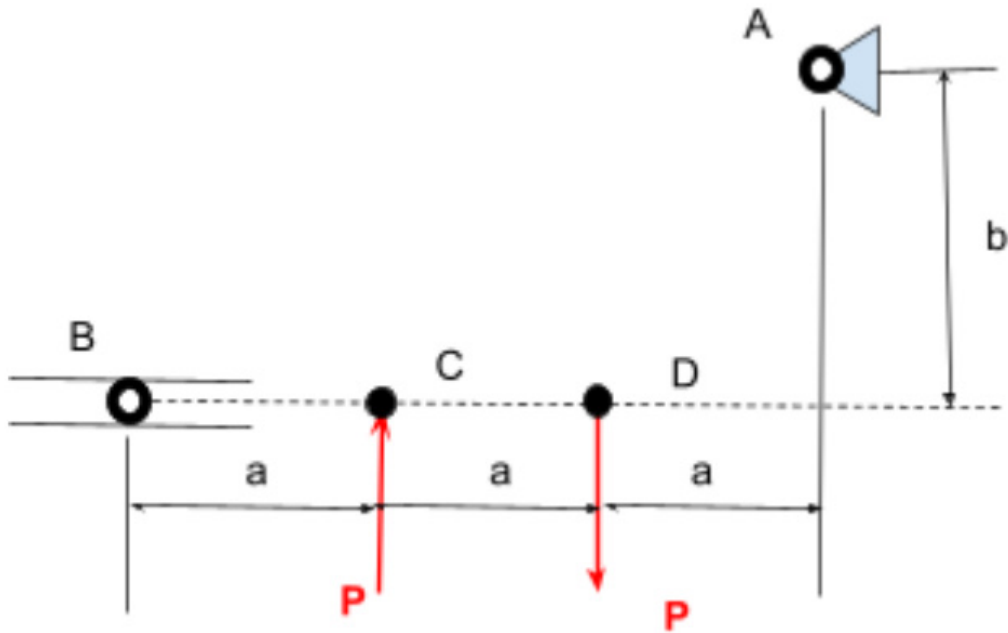


A2 Design with Basic Stresses

Zechariah
(Zac)
Diegelman

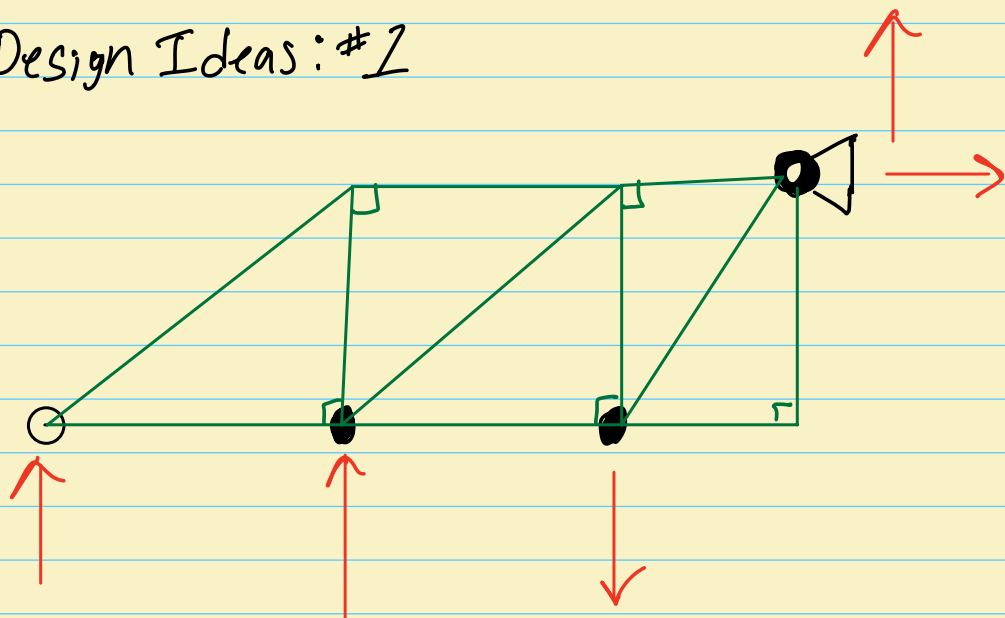
Objectives:

- Design Lightweight Planar Truss (A500 Steel)
 - ↳ Design Constraints shown figure below.
 - ↳ Cross Sectional Area of each element is to be identical
 - ↳ Identical Pins, every element has same cross-sec Area

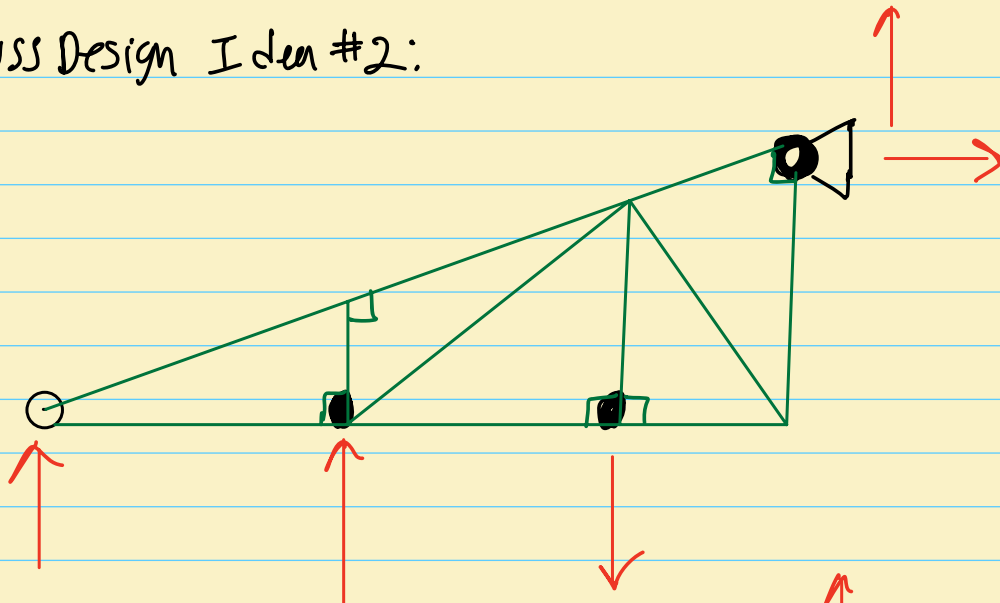


GIVEN: $P = 15 \text{ kN}$ $a = 0.4 \text{ m}$ $A = \text{Pin}$
 $b = 0.3 \text{ m}$ $B = \text{Roller}$

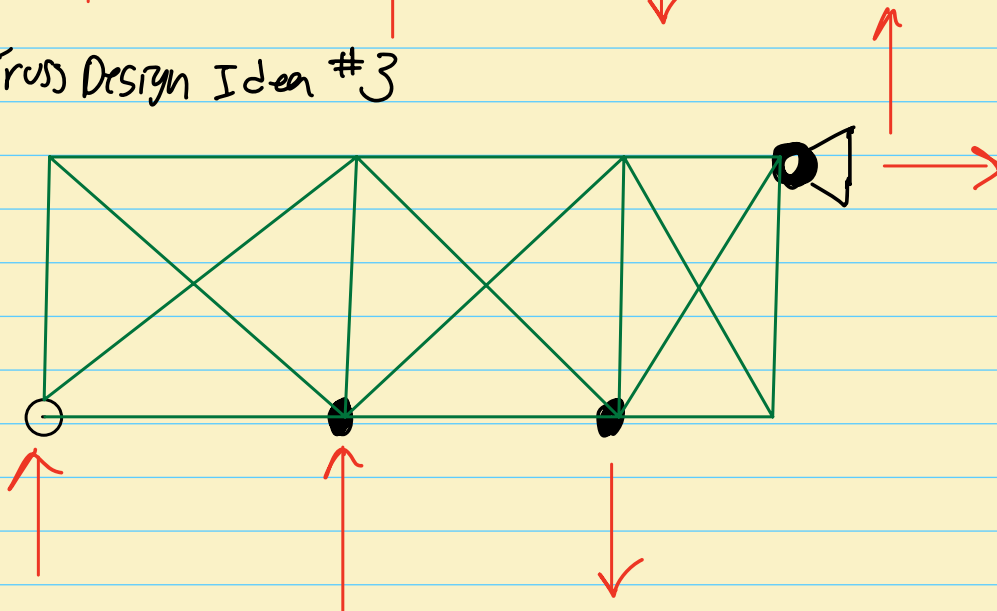
Truss Design Ideas: #1



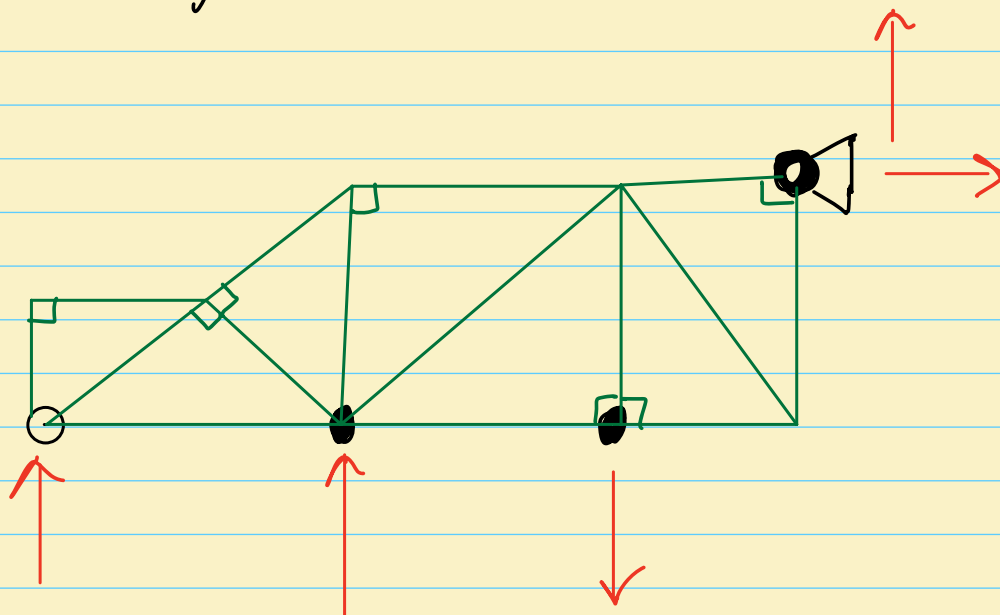
Truss Design Idea #2:



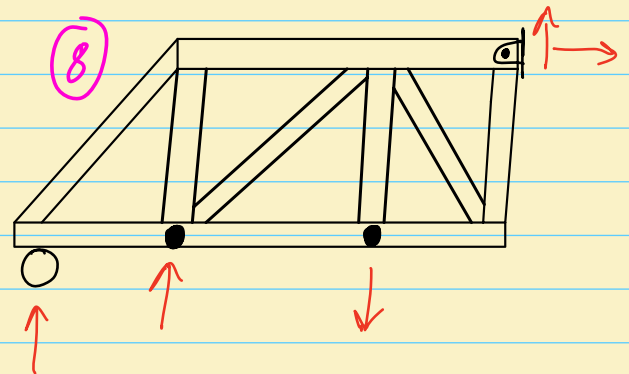
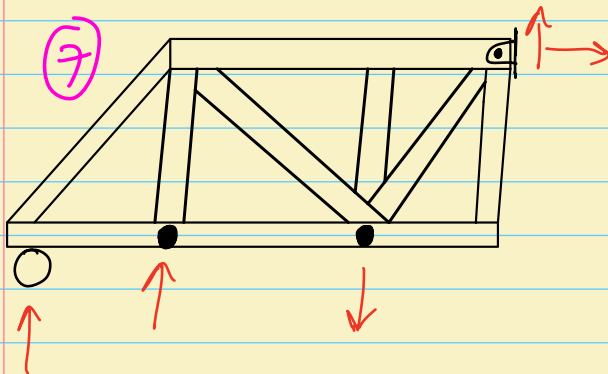
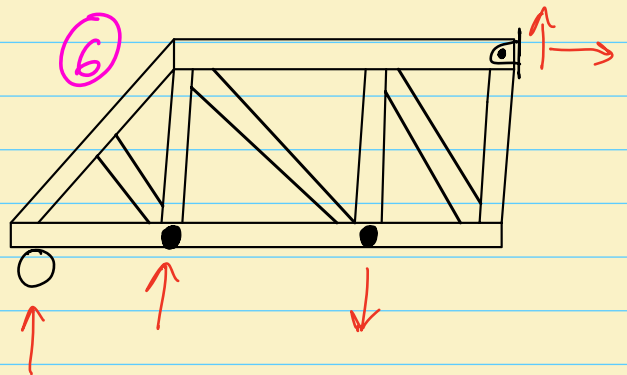
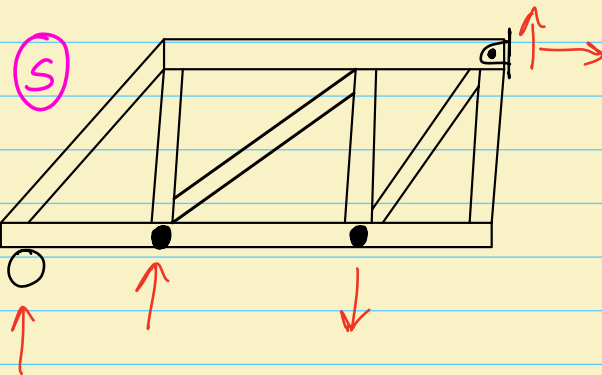
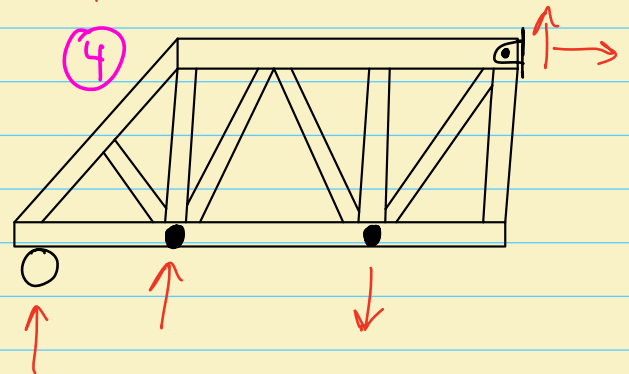
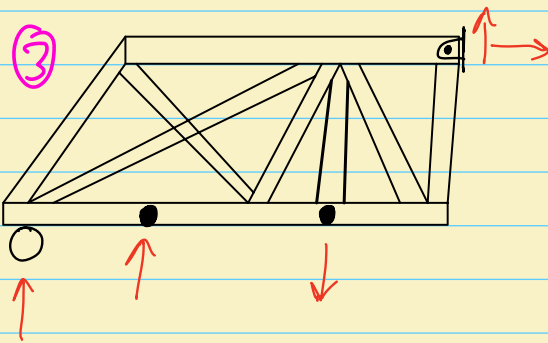
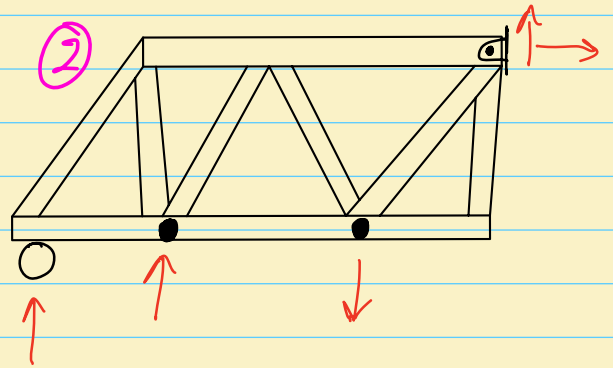
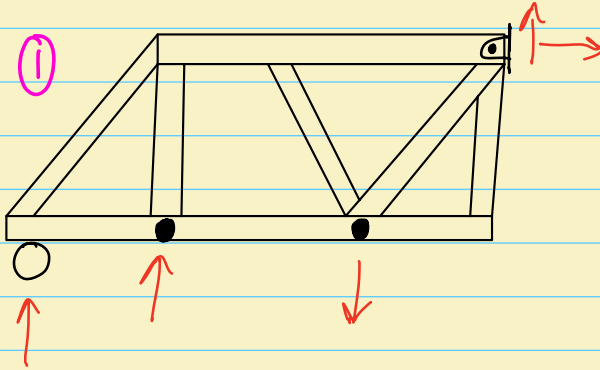
Truss Design Idea #3



Truss Design Idea #4

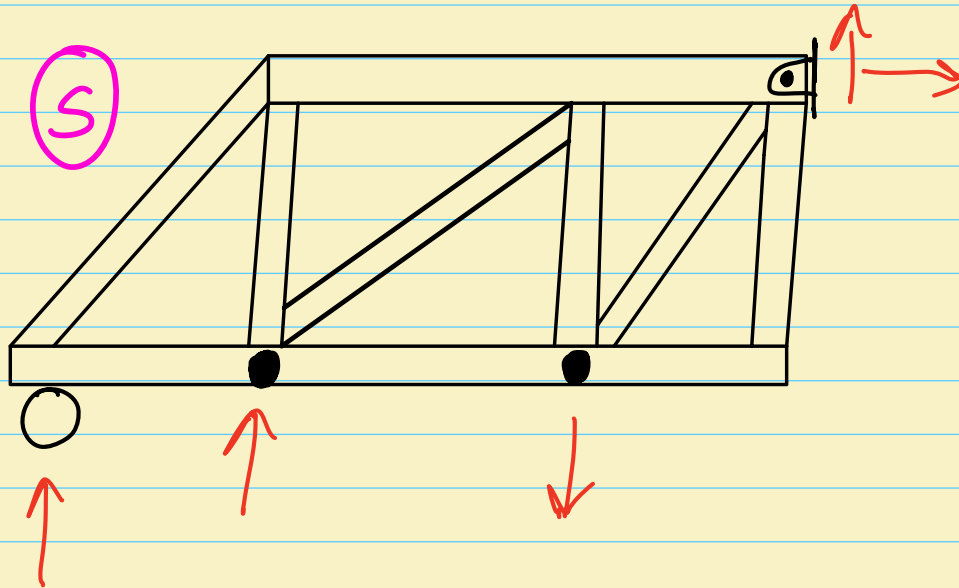


More Design Ideas

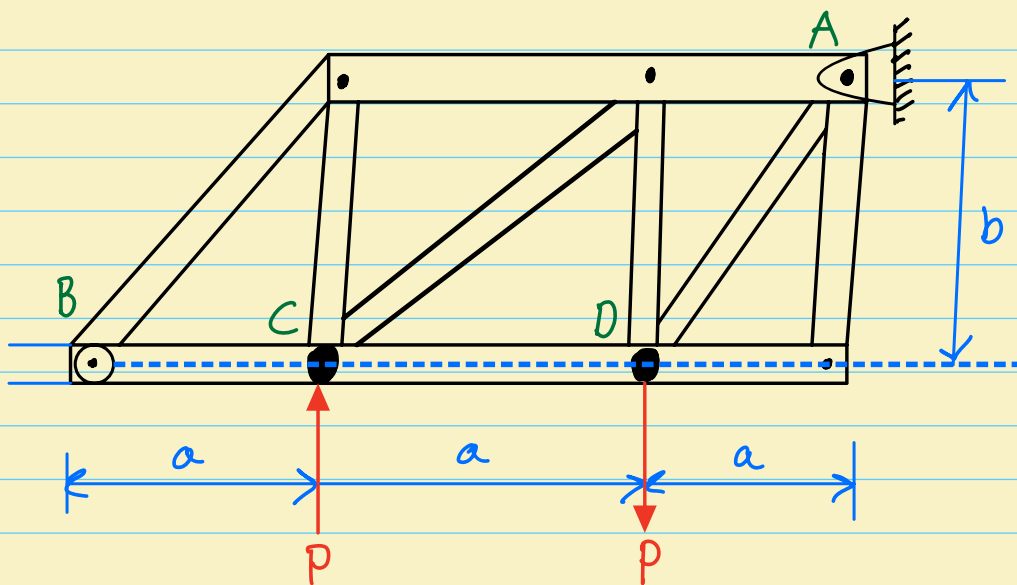


Design

I decided #5 would be the most practicle choice, for design, and overall weight.



Now to add the Parameters:



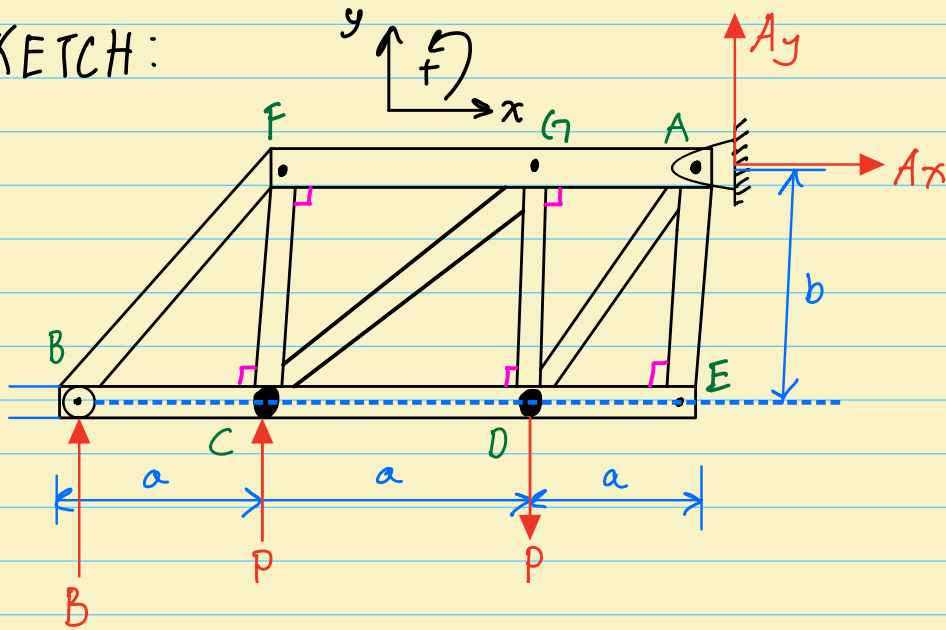
GIVEN:

$$P = 15 \text{ kN} \quad a = 0.4 \text{ m} \quad A = \text{Pin}$$

$$b = 0.3 \text{ m} \quad B = \text{Roller}$$

FBD: Chosen Design

SKETCH:

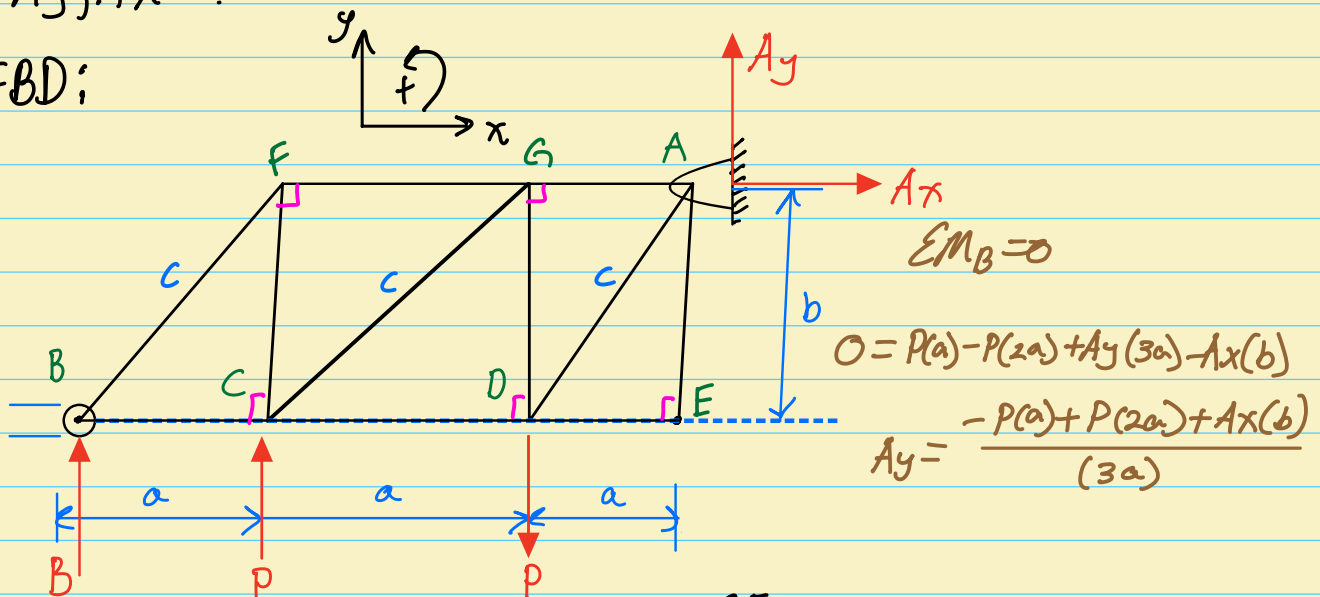


GIVEN:

$$P = 15 \text{ kN} \quad a = 0.4 \text{ m} \quad A = \text{Pin} \\ b = 0.3 \text{ m} \quad B = \text{Roller}$$

FIND: $B = ?$
 $A_y, A_x = ?$

FBD:



$$\sum M_A = 0$$

$$0 = P(a) - P(2a) - B(3a)$$

$$B = \frac{P(a) - P(2a)}{(3a)}$$

$$\sum F_y = 0$$

$$0 = B + A_y + P - P$$

$$A_y = P - P - B$$

$$\sum F_x = 0$$

$$A_x = 0$$

Plug in Numbers

$$B = \frac{P(a) - P(2a)}{(3a)}$$

$$B = \frac{15\text{KN}(0.4\text{m}) - 15\text{KN}(2(0.4\text{m}))}{(3(0.4\text{m}))}$$

$$B = -5\text{KN (Pull)}$$

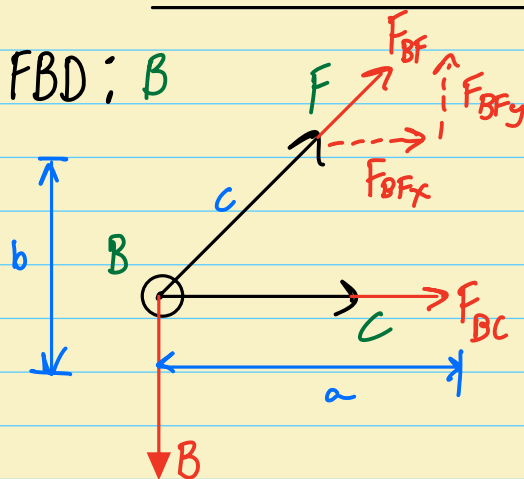
$$A_y = P - P - B$$

$$A_y = 15 - 15 - (-5)$$

$$A_y = 5\text{KN}$$

$$A_x = 0\text{KN}$$

Method Of Joints



$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{(0.4\text{m})^2 + (0.3\text{m})^2}$$

$$c = 0.5\text{m}$$

messed up big time with F_{BC} , I didn't subtract the two originally when doing the $F_x = 0$ & said it was in tension,

$$\sin \theta = \frac{b}{c}$$

$$\sin \theta = \frac{0.3\text{m}}{0.5\text{m}}$$

$$\cos \theta = \frac{a}{c}$$

$$\cos \theta = \frac{0.4\text{m}}{0.5\text{m}}$$

$$\sum F_y = 0$$

$$0 = F_{BF} \sin \theta - B$$

$$0 = F_{BF} \left(\frac{0.3}{0.5} \right) - B$$

$$F_{BF} = \frac{B}{\left(\frac{0.3}{0.5} \right)}$$

$$F_{BF} = 8.33\text{KN (T)}$$

$$\sum F_x = 0$$

$$0 = F_{BFx} + F_{BC}$$

$$F_{BC} = -F_{BF} \left(\frac{0.4}{0.5} \right)$$

$$F_{BC} = -8.33\text{KN} \left(\frac{0.4}{0.5} \right)$$

$$F_{BC} = -6.66\text{KN (C)}$$

$$F_{BF} = 8.33\text{KN (T)}$$

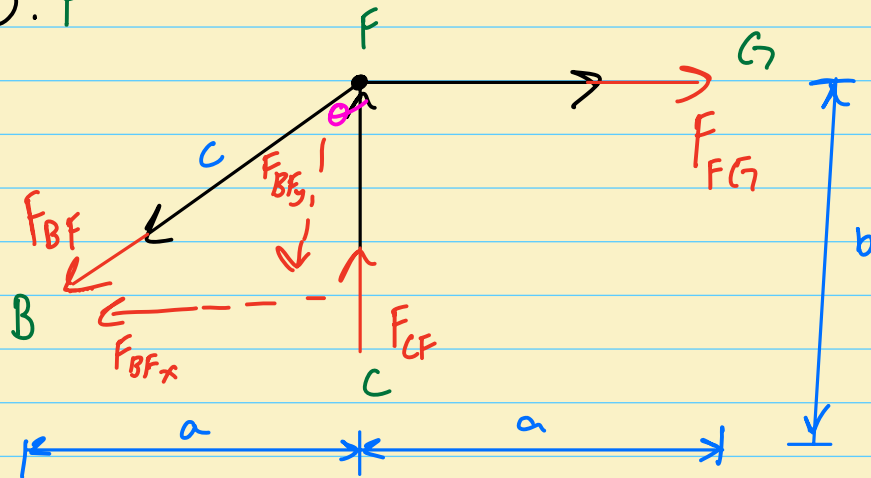
$$F_{BC} = 6.66\text{KN (C)}$$

had to redo my problem when

Method of Joints

evaluating for joint A.

FBD: F



$$\sin \theta = \left(\frac{a}{c} \right)$$

$$\sin \theta = \left(\frac{0.4}{0.5} \right)$$

$$\cos \theta = \left(\frac{b}{c} \right)$$

$$\cos \theta = \left(\frac{0.3}{0.5} \right)$$

KNOWN: $F_{BF} = 8.33 \text{ KN}$, $c = 0.5 \text{ m}$, $a = 0.4 \text{ m}$, $b = 0.3 \text{ m}$

$$\sum F_x = 0$$

$$0 = F_{FG} - F_{BF} \cos \theta$$

$$F_{FG} = F_{BF} \cos \theta$$

$$F_{FG} = 8.33 \text{ KN} \left(\frac{0.3}{0.5} \right)$$

$$F_{FG} = 5 \text{ KN (T)}$$

$$\sum F_y = 0$$

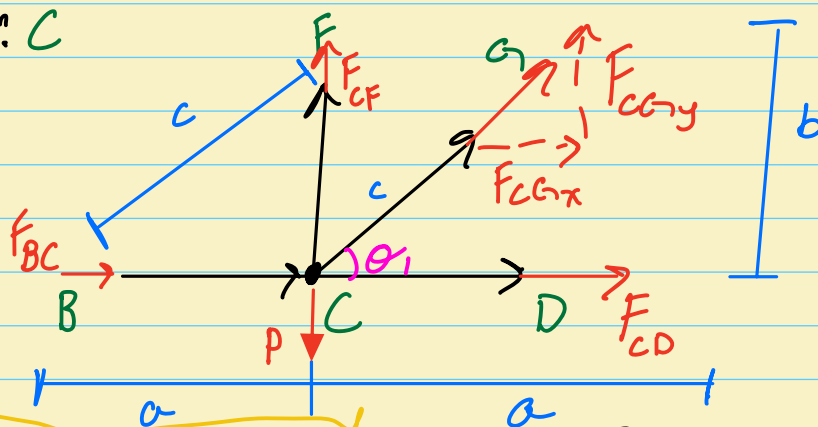
$$0 = F_{CF} - F_{BF} \sin \theta$$

$$F_{CF} = F_{BF} \sin \theta$$

$$F_{CF} = 8.33 \text{ KN} \left(\frac{0.4}{0.5} \right)$$

$$F_{CF} = 6.666 \text{ KN (C)}$$

FBD: C



$$\sin \theta = \left(\frac{b}{c} \right)$$

$$\sin \theta = \left(\frac{0.3}{0.5} \right)$$

$$\cos \theta = \left(\frac{a}{c} \right)$$

$$\cos \theta = \left(\frac{0.4}{0.5} \right)$$

KNOWN: $F_{BC} = 6.666 \text{ KN (C)}$, $P = 15 \text{ KN}$, $F_{CF} = 6.666 \text{ KN}$,
 $a = 0.4 \text{ m}$, $b = 0.3 \text{ m}$, $c = 0.5 \text{ m}$

$$\sum F_x = 0$$

$$0 = F_{CD} + F_{BC} + F_{CG} \cos \theta$$

$$F_{CD} = -F_{BC} - F_{CG} \cos \theta$$

$$\sum F_y = 0$$

$$0 = F_{CF} + F_{CG} \sin \theta - P$$

$$F_{CG} = \frac{P - F_{CF}}{\sin \theta}$$

My mess up
 at joint
 B make me
 continue to
 use wrong
 values &
 realize

drawn

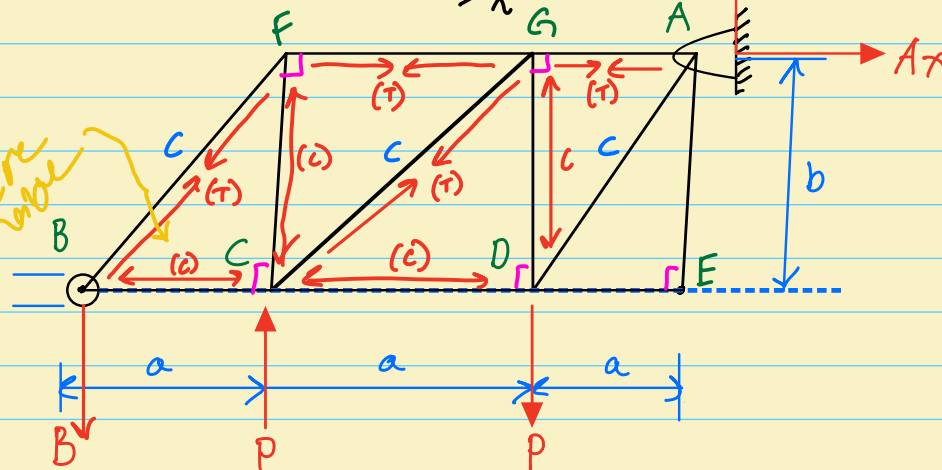
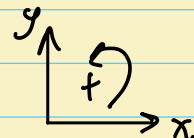
$$F_{CD} = -6.66 \bar{K}N - 13.89 \bar{K}N \left(\frac{0.4}{0.5} \right) \quad F_{CG} = \frac{15 \bar{K}N - 6.66 \bar{K}N}{\left(\frac{0.3}{0.5} \right)}$$

$$F_{CD} = -4.45 \bar{K}N (C)$$

$$F_{CG} = 13.89 \bar{K}N (T)$$

$$F_{CD} = -17.78 \bar{K}N (C)$$

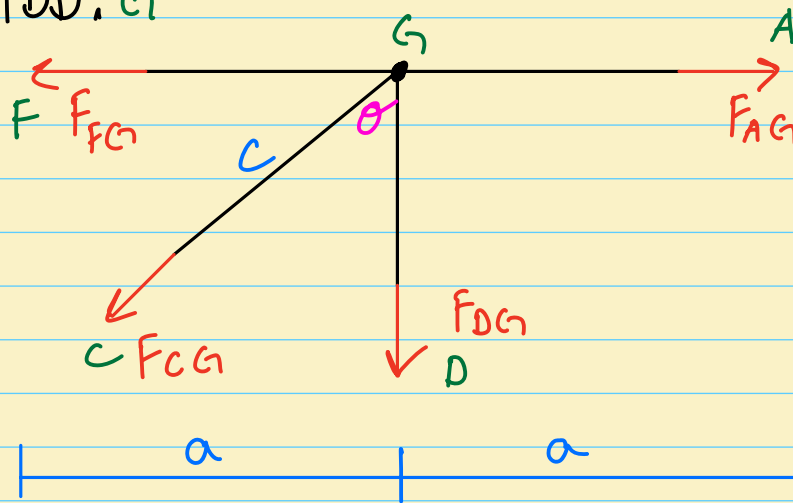
Reevaluating
(Visualizing)



even misse I
it was to draw

Method of Joints

FBD: G



$$\sin \theta = \left(\frac{a}{c} \right)$$

$$\sin \theta = \left(\frac{0.4}{0.5} \right)$$

$$\cos \theta = \left(\frac{b}{c} \right)$$

$$\cos \theta = \left(\frac{0.3}{0.5} \right)$$

Known: $F_{FG} = 5 \bar{K}N$, $F_{CG} = 13.89 \bar{K}N$, $a = 0.4 \text{ m}$, $b = 0.3 \text{ m}$, $c = 0.5 \text{ m}$

$$\sum F_y = 0$$

$$0 = -F_{CG} \cos \theta - F_{DG}$$

$$F_{DG} = -F_{CG} \cos \theta$$

$$F_{DG} = -8.334 \bar{K}N (C)$$

$$\sum F_x = 0$$

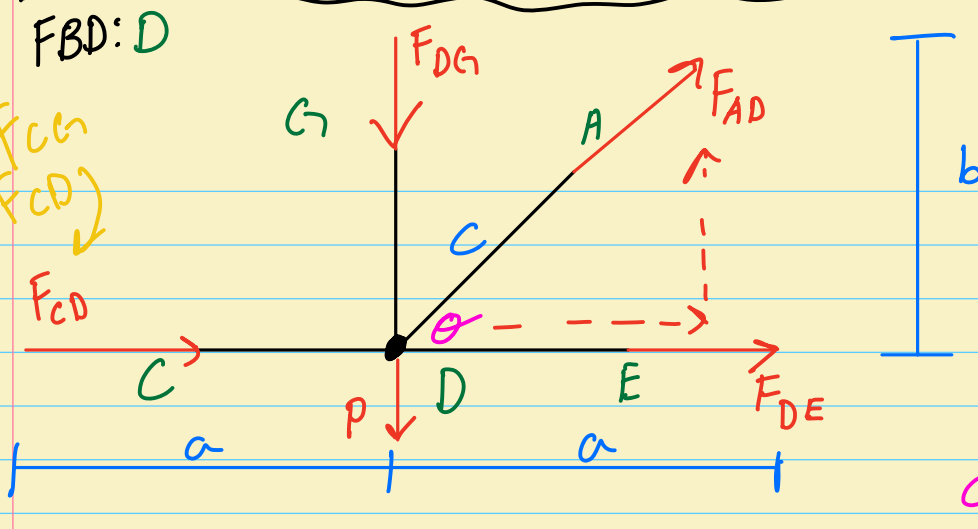
$$0 = F_{AG} - F_{FG} - F_{CG} \sin \theta$$

$$F_{AG} = F_{FG} + F_{CG} \sin \theta$$

$$F_{AG} = 5 \bar{K}N + 13.89 \bar{K}N \left(\frac{0.4}{0.5} \right)$$

$$F_{AG} = 16.11 \bar{K}N (T)$$

Accidentally
called this F_{CD}
instead of F_{CD}



KNOWN: $F_{CD} = -17.78 \text{ kN (C)}$, $F_{DG} = 8.34 \text{ kN (C)}$, $P = 1 \text{ kN}$,
 $a = 0.4 \text{ m}$, $b = 0.3 \text{ m}$, $c = 0.5 \text{ m}$

$$\sum F_y = 0$$

$$0 = F_{AD} \sin \theta - P - F_{DG}$$

$$F_{AD} = \frac{P + F_{DG}}{\sin \theta}$$

$$F_{AD} = \frac{(1 \text{ kN}) + (8.34 \text{ kN})}{\left(\frac{0.3}{0.5}\right)}$$

$$F_{AD} = 38.89 \text{ kN}$$

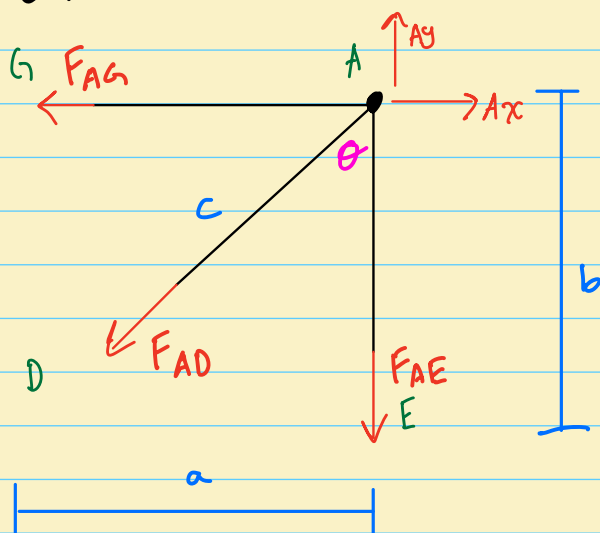
$$\sum F_x = 0$$

$$0 = F_{AD} \cos \theta + F_{CD} + F_{DE}$$

$$F_{DE} = -F_{AD} \cos \theta - F_{CD}$$

$$F_{DE} = 38.89 \text{ kN} \left(\frac{0.4}{0.5}\right) - (-17.78 \text{ kN})$$

FBD: A

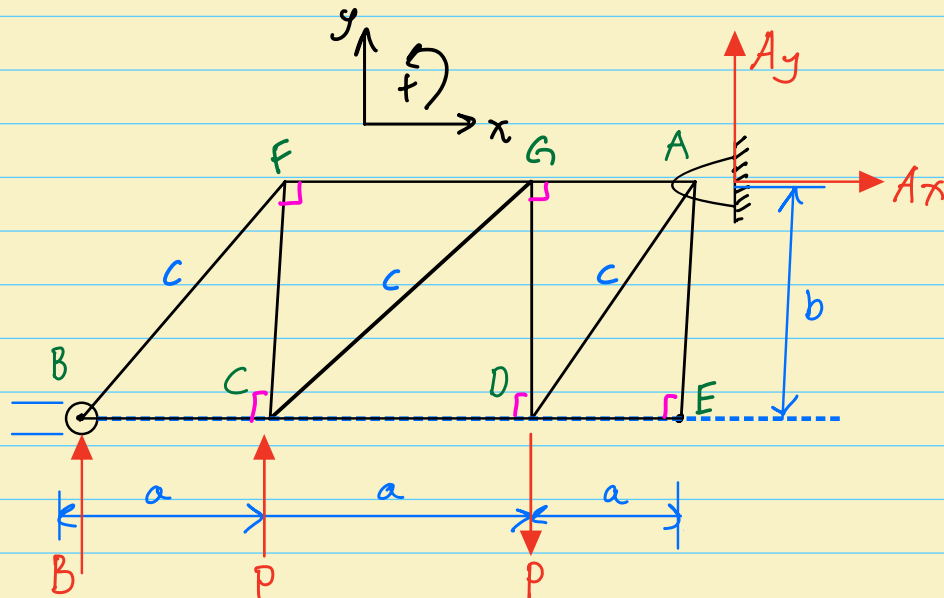


KNOWN: $F_{AG} = 16.11 \text{ kN}$, F_{AE} , $A_y = 5 \text{ kN}$, $A_x = 0 \text{ kN}$
 $a = 0.4 \text{ m}$, $b = 0.3 \text{ m}$, $c = 0.5 \text{ m}$

$$\sum F_x = 0$$

0 -

Felt Like I messed up
Somewhere, decided to redo it.



KNOWN: $P = 1 \text{ kN}$

$$a = 0.4 \text{ m}$$

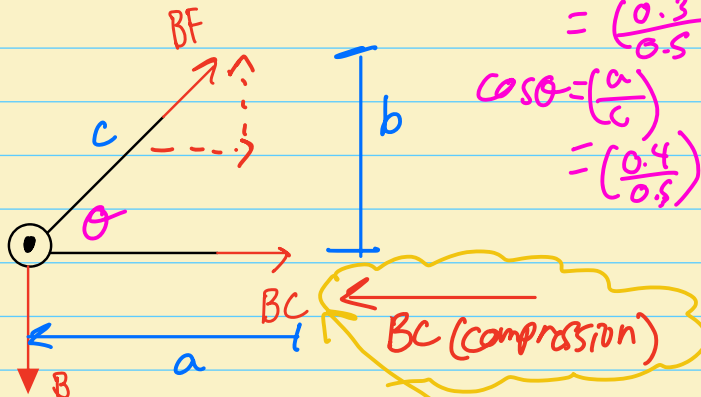
$$B = -1 \text{ kN}$$

$$b = 0.3 \text{ m}$$

$$A_y = 1 \text{ kN}$$

$$A_x = 0$$

FBD: Joint B



$$\sin \theta = \left(\frac{b}{c} \right) = \left(\frac{0.3}{0.5} \right)$$

$$\cos \theta = \left(\frac{a}{c} \right) = \left(\frac{0.4}{0.5} \right)$$

$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{0.4^2 + 0.3^2}$$

$$c = 0.5 \text{ m}$$

$$\sum F_x = 0$$

$$0 = BC + BF \cos \theta$$

$$\sum F_y = 0$$

$$0 = BF \sin \theta - B$$

$$BF = \frac{B}{\sin \theta}$$

$$BF = \frac{5 \text{ kN}}{\left(\frac{0.3}{0.5}\right)}$$

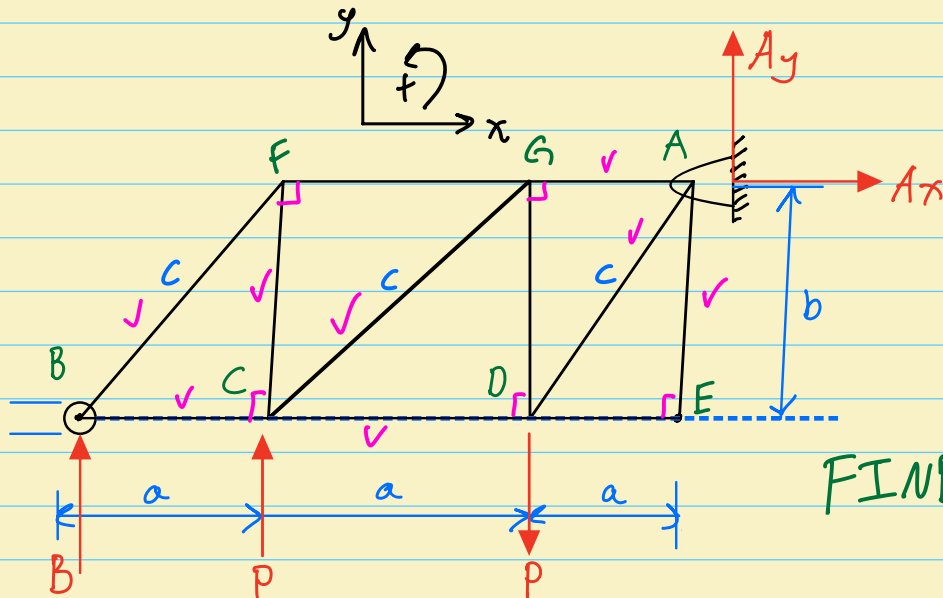
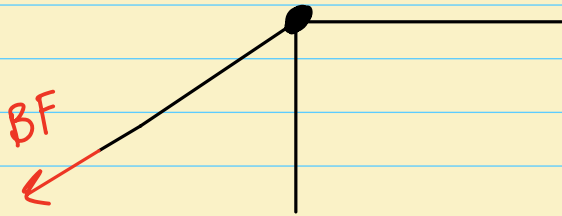
$$BF = 8.34 \text{ kN (T)}$$

$$BC = -BF \cos \theta$$

$$BC = -8.34 \text{ kN} \left(\frac{0.4}{0.5}\right)$$

$$BC = 6.67 \text{ kN (C)}$$

FBD: Joint F



FIND: AE =

AD =

AG =

BF =

BC =

CF =

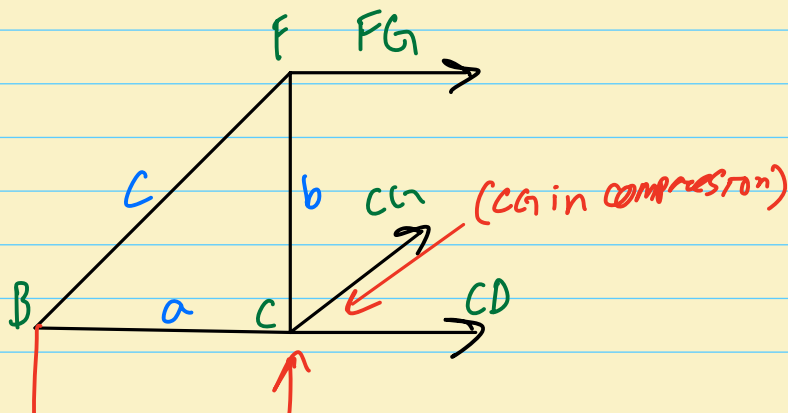
CG =

CD =

FG =

DG =

DE =



$$\downarrow B = 5 \text{ kN} \quad \uparrow P = 15 \text{ kN}$$

$$\sum M_C = 0$$

$$0 = B(a) - F_G(b)$$

$$F_G = B(a)$$

$$F_G = 5 \text{ kN} (0.4 \text{ m})$$

$$F_G = 2 \text{ kN (T)}$$

$$\sum F_y = 0$$

$$0 = -B + P + C_G \sin \theta$$

$$C_G = \frac{B - P}{\sin \theta}$$

$$C_G = \frac{5 \text{ kN} - 15 \text{ kN}}{\left(\frac{0.3}{0.5}\right)}$$

$$\sum F_x = 0$$

$$0 = F_G + C_D - C_G \cos \theta$$

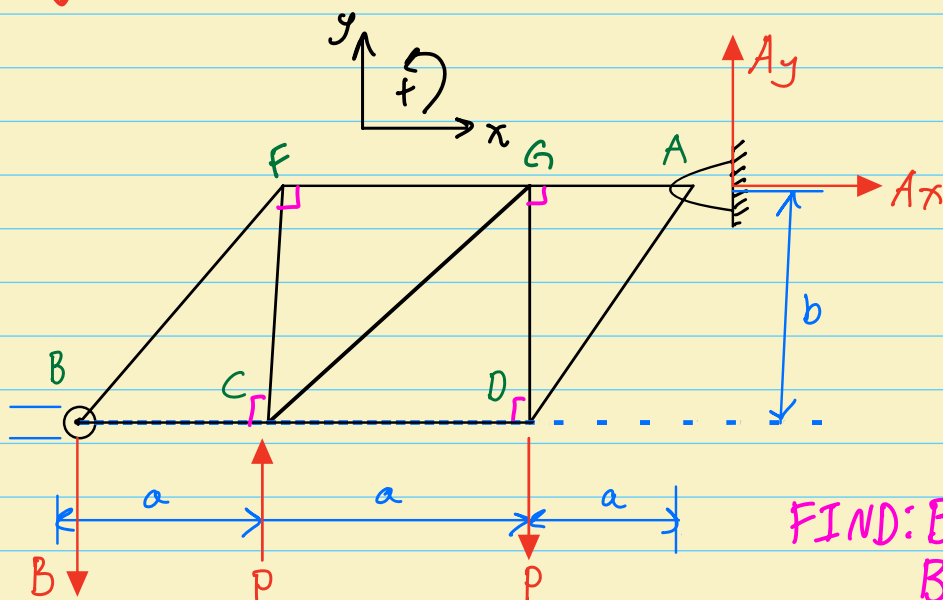
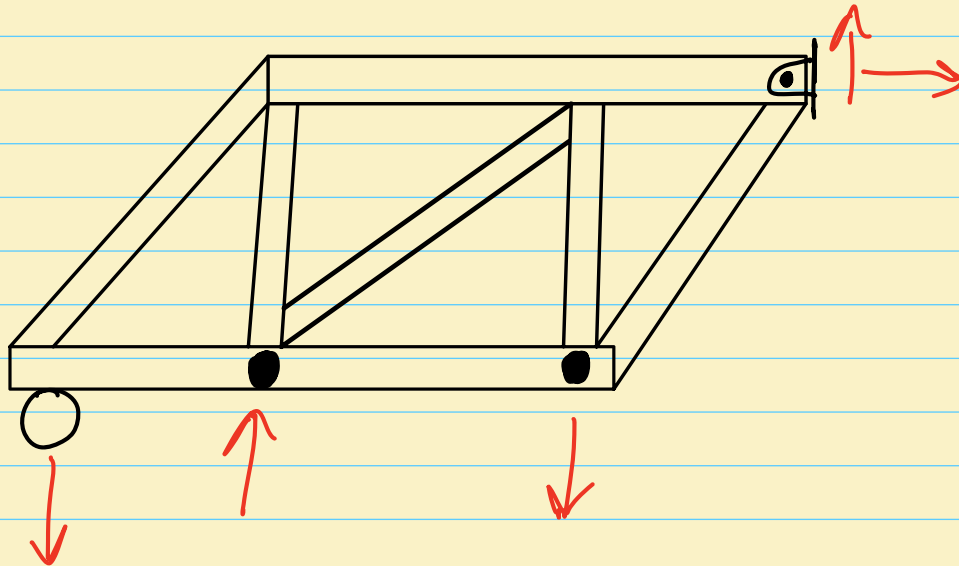
$$C_G = 16.67 \text{ kN (C)}$$

$$C_D = C_G \cos \theta - F_G$$

$$C_D = 16.67 \text{ kN} \left(\frac{0.4}{0.5}\right) - 2 \text{ kN}$$

$$C_D = 11.34 \text{ kN (T)}$$

New Plan



FIND: $BF = \checkmark$

$BC = \checkmark$

$CF = \checkmark$

$FG = \checkmark$

$CG = \checkmark$

$CD = \checkmark$

$AG = \checkmark$

$DG = \checkmark$

$AD = \checkmark$

Known: $B = 5 \text{ kN}$

$A_x = 0$

$A_y = 5 \text{ kN}$

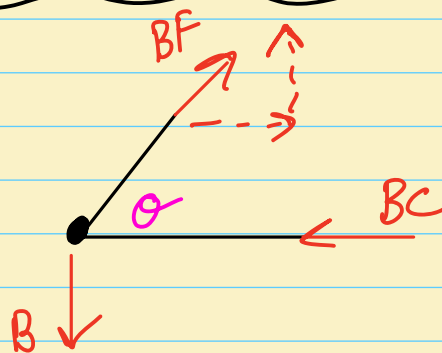
$P = 15 \text{ kN}$

$a = 0.4 \text{ m}$

$b = 0.3 \text{ m}$

$c = 0.5 \text{ m}$

Joint B:



$\sum F_y = 0$

$$0 = BF \sin \theta - B$$

$$BF = \frac{B}{\sin \theta}$$

$$BF = \frac{5}{\sin(36.87^\circ)}$$

$$BF = 8.33 \text{ kN (T)}$$

$$\sum F_x = 0$$

$$0 = -BC + BF \cos \theta$$

$$BC = BF \cos \theta$$

$$BC = 8.33 \text{ KN} \cos(36.87^\circ)$$

$$BC = 6.67 \text{ KN (C)}$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

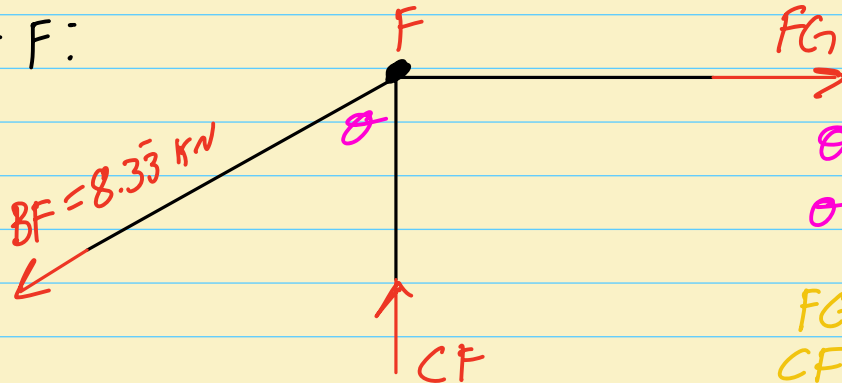
$$\theta = \tan^{-1}\left(\frac{0.3}{0.4}\right)$$

$$\theta = 36.87^\circ$$

$$BF = 8.34 \text{ KN (T)}$$

$$BC = 6.67 \text{ KN (C)}$$

Joint F:



$$\theta = \tan^{-1}\left(\frac{a}{b}\right)$$

$$\theta = 53.13^\circ$$

$$FG = 6.67 \text{ KN (T)}$$

$$CF = 5 \text{ KN (T)}$$

$$\sum F_x = 0$$

$$0 = FG - BF \sin \theta$$

$$FG = 8.33 \text{ KN} \sin(53.13^\circ)$$

$$FG = 6.67 \text{ KN (T)}$$

$$\sum F_y = 0$$

$$0 = CF - BF \cos \theta$$

$$CF = BF \cos \theta$$

$$CF = 8.33 \text{ KN} \cos(53.13^\circ)$$

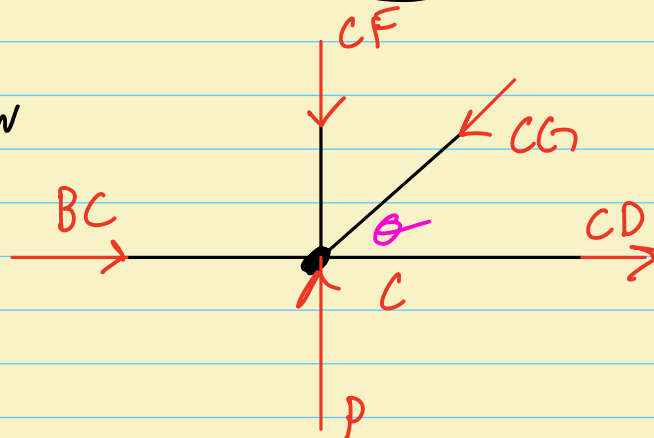
$$CF = 5 \text{ KN (C)}$$

Joint C:

$$BC = 6.67 \text{ KN}$$

$$CF = 5 \text{ KN}$$

$$P = 15 \text{ KN}$$



$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\theta = 36.87^\circ$$

$$CG = 16.67 \text{ KN (C)}$$

$$CD = 6.67 \text{ KN (T)}$$

$$\sum F_y = 0$$

$$0 = -CF + P - CG \sin \theta$$

$$CG = \frac{-CF + P}{\sin \theta}$$

$$CG = \frac{-5 \text{ KN} + 15 \text{ KN}}{\sin(36.87^\circ)}$$

$$CG = 16.67 \text{ KN (C)}$$

$$\sum F_x = 0$$

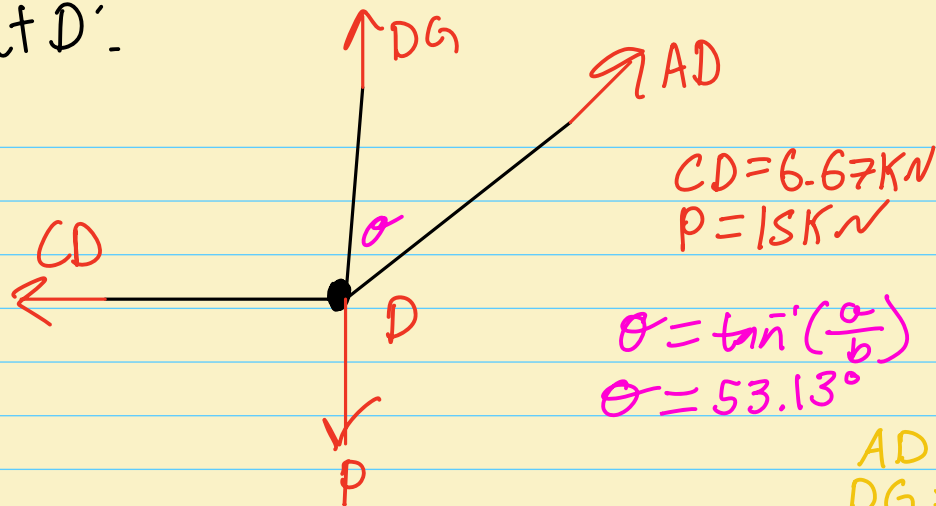
$$0 = BC + CD - CG \cos \theta$$

$$CD = -BC + CG \cos \theta$$

$$CD = -6.67 \text{ KN} + 16.67 \text{ KN} \cos(36.87^\circ)$$

$$CD = 6.67 \text{ KN (T)}$$

Joint D:



$$AD = 8.34 \text{ kN (T)}$$

$$DG = 10 \text{ kN (T)}$$

$$\sum F_x = 0$$

$$0 = AD \sin \theta - CD$$

$$AD = \frac{CD}{\sin \theta}$$

$$AD = 8.34 \text{ kN (T)}$$

$$\sum F_y = 0$$

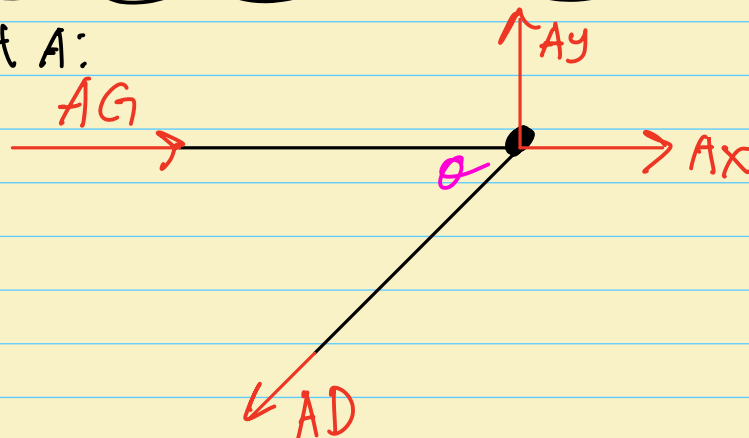
$$0 = DG - P + AD \cos \theta$$

$$DG = P - AD \cos \theta$$

$$DG = 15 \text{ kN} - 8.34 \text{ kN} \cos(53.13^\circ)$$

$$DG = 10 \text{ kN (T)}$$

Joint A:



$$Ay = 5 \text{ kN}$$

$$Ax = 0$$

$$AD = 8.34 \text{ kN (T)}$$

$$\sum F_x = 0$$

$$0 = Ax + AG - AD \cos \theta$$

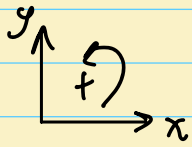
$$AG = Ax + AD \cos \theta$$

$$AG = 0 + 8.34 \text{ kN} \cos(36.87^\circ)$$

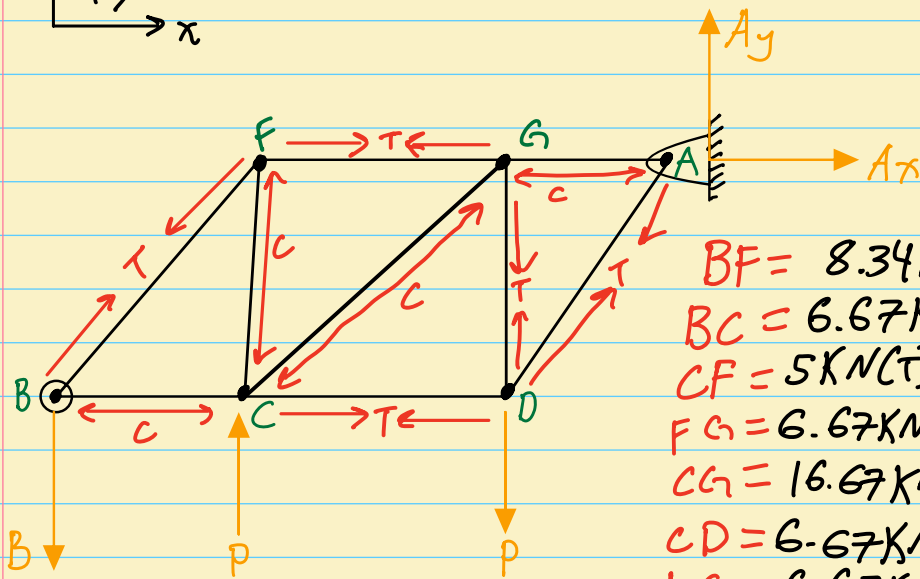
$$AG = 6.67 \text{ kN (C)}$$

$$AG = 6.67 \text{ kN (C)}$$

All Internal & Reaction Forces Known



Evaluate:



$BF = 8.34 \text{ kN (T)}$
 $BC = 6.67 \text{ kN (C)}$
 $CF = 5 \text{ kN (T)}$
 $FG = 6.67 \text{ kN (T)}$
 $CG = 16.67 \text{ kN (C)}$
 $CD = 6.67 \text{ kN (T)}$
 $AG = 6.67 \text{ kN (C)}$
 $DG = 10 \text{ kN (T)}$
 $AD = 8.34 \text{ kN (T)}$
 $P = 15 \text{ kN UP/Down}$
 $B = 5 \text{ kN Downward}$
 $A_y = 5 \text{ kN Upward}$
 $A_x = 0$

Largest force = CG
 force of 16.67 kN

Next steps: find stresses

FIND: $\sigma_{\max} = ?$ $A = ?$ weight = ?

GIVEN: Material A500 structural steel
 $SF = 3.5$ $S_y = 315 \text{ MPa}$

EQ's:

$$\sigma = \frac{F}{A}$$

$$\frac{S_y}{SF} > \sigma$$

$$A_{\min} > \frac{F(SF)}{S_y}$$

$$\frac{S_y}{SF} > \frac{F_{\text{largest}}}{A}$$

$$\frac{S_y}{SF} > \frac{V}{A}$$

$$\frac{S_y}{SF} > \sigma$$

$$\frac{315 \text{ MPa}}{3.5} > \sigma$$

$$90 \text{ MPa} > \sigma$$

$$A = \frac{F_{\max} (SF)}{S_y}$$

$$A = \frac{16.67 \text{ kN} (3.5)}{315 \text{ MPa}}$$

$$A = \frac{16,666.67 \text{ N} (3.5)}{315 \times 10^6 \text{ N/m}^2}$$

$$A = 1.85 \times 10^{-4} \text{ m}^2$$

$P = \text{Density}$

$$P = 7800 \text{ kg/m}^3$$

$$\text{weight} = P \cdot A \cdot L_{\text{total}}$$

$$\text{weight} = P \cdot A \cdot (2b + 3c + 4a)$$

$$W = 7800 \text{ kg/m}^3 \cdot (1.85 \times 10^{-4} \text{ m}^2) (2(0.3\text{m}) + 3(0.5\text{m}) + 4(0.4\text{m}))$$

$$W = 5.34 \text{ kg} \left(\frac{2.21 \text{ lb}}{1 \text{ kg}} \right) = 11.77 \text{ lb}$$

$$A = 1.85 \times 10^{-4} \text{ m}^2$$

$$\text{width} = \sqrt{1.85 \times 10^{-4} \text{ m}^2}$$

$$\text{width} = 0.0136 \text{ m}$$

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{16.67 \text{ kN}}{1.85 \times 10^{-4} \text{ m}^2}$$

$$\sigma = 90 \times 10^3 \text{ MPa}$$

Cross Sectional Area Pins

- Determine x-sectional area of connecting Pins
- Made of hardened tool steel
- Yield Shear Strength = 170 Ksi
- P (density) = 0.278 lb/in³
- Calc required x-sect area of pins to withstand expected Shear forces.
- Design a Single Shear connection
- Use SF of 4

Known: For Connecting Pins

$$SF = 4$$

$$S_y = 170 \text{ Ksi}$$

$$P = 0.287 \text{ lb/in}^3$$

For Planer Truss

$$A = 1.85 \times 10^4 \text{ m}^2$$

$$W = 5.34 \text{ Kg}$$

$$BF = 8.34 \text{ KN (T)}$$

$$BC = 6.67 \text{ KN (C)}$$

$$CF = 5 \text{ KN (T)}$$

$$FG = 6.67 \text{ KN (T)}$$

$$CG = 16.67 \text{ KN (C)}$$

$$CD = 6.67 \text{ KN (T)}$$

$$AG = 6.67 \text{ KN (C)}$$

$$DG = 10 \text{ KN (T)}$$

$$AD = 8.34 \text{ KN (T)}$$

$$P = 15 \text{ KN UP/Down}$$

$$B = 5 \text{ KN Downward}$$

$$A_y = 5 \text{ KN Upward}$$

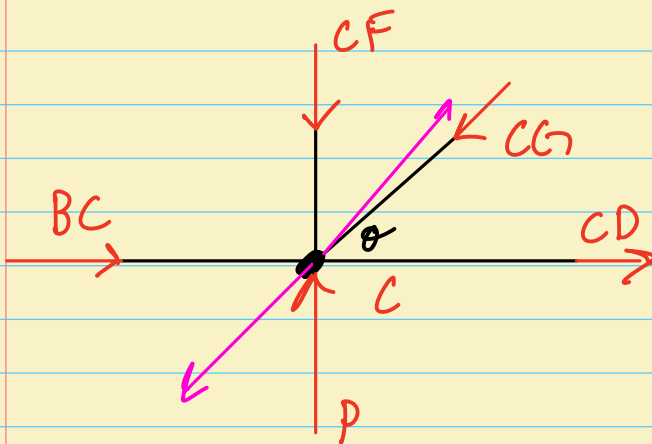
$$A_x = 0$$

Unknown: $A_{pin} = ?$
 $W_{total} = ?$

Formulas:

- $\text{weight} = P \cdot A \cdot L_{\text{total}}$
- $\tau = \frac{V}{A}$
- $A = \frac{F_{\text{max}} (SF)}{S_y}$
- $A_{\text{min}} > \frac{F (SF)}{S_y}$
- $\frac{S_y}{SF} > \sigma$
- $\frac{S_y}{SF} > \frac{V}{A}$
- $\frac{S_y}{SF} > \frac{F_{\text{max}}}{A}$

FBD: Pin Single Shear



$$\begin{aligned} CF &= 5 \text{ kN} & SF &= 4 \\ CG &= 16.67 \text{ kN} & S_y &= 170 \text{ ksi} \\ BC &= 6.67 \text{ kN} & P &= 0.287 \text{ lb/in}^3 \\ CD &= 6.67 \text{ kN} \\ P &= 15 \text{ kN} \end{aligned}$$

$$\bullet A = \frac{F_{\max} (SF)}{S_y} \quad \bullet \tau = \frac{V}{A}$$

$$170 \times 10^3 = 170,000 \frac{\text{lb}}{\text{in}^2} \left(\frac{4.45 \text{ N}}{2 \text{ lb}} \right) \left(\frac{2 \text{ in}^2}{6.45 \text{ mm}^2} \right) = 117257.73 \frac{\text{N}}{\text{mm}^2}$$

$$\begin{aligned} 2 \text{ in} &= 25.4 \text{ mm} \\ 2 \text{ in}^2 &= (25.4 \text{ mm})^2 \\ &= 645 \text{ mm}^2 \end{aligned}$$

$$A = \frac{16.67 \times 10^3 \text{ N} (4)}{117,257.73 \text{ N/mm}^2}$$

$$A = 0.57 \text{ mm}^2$$

$$L = 0.036 \text{ m} \left(\frac{39.37 \text{ in}}{1 \text{ m}} \right) = 0.535 \text{ in}$$

$$\tau = \frac{16,670 \text{ N}}{0.57 \text{ mm}^2}$$

$$\tau = 29,314.43 \text{ MPa}$$

$$0.57 \text{ mm}^2 \left(\frac{\text{in}^2}{645.16 \text{ mm}^2} \right) =$$

$$P = 0.287 \text{ lb/in}^3$$

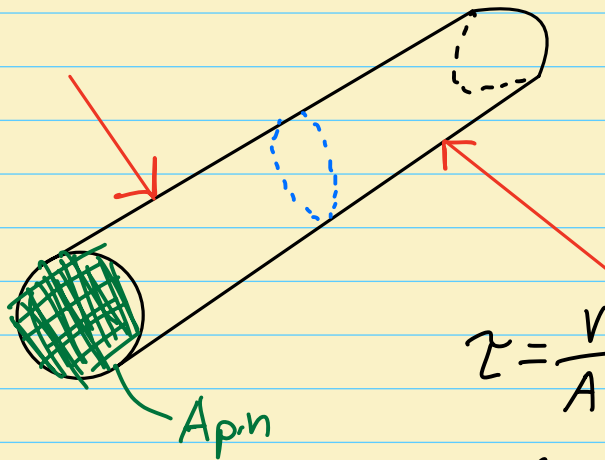
$$\rightarrow = 8.83 \times 10^{-4} \text{ in}^2$$

$$W = P \cdot A \cdot L_{\text{total}}$$

$$W = 0.287 \text{ lb/in}^3 (8.83 \times 10^{-4} \text{ in}^2) (0.535 \text{ in} (6 \text{ pins}))$$

$$W = 8.135 \times 10^{-4} \text{ lb} \left(\frac{1 \text{ kg}}{2.2 \text{ lb}} \right) = 3.69 \times 10^{-4} \text{ kg}$$

FBD: Connecting Pin



$$\tau = 29,314.43 \text{ MPa}$$

$$A = 0.57 \text{ mm}^2$$

$$\tau = \frac{V}{A}$$

$$A = \frac{V}{\tau}$$

$$A = \frac{16.67 \text{ kN}}{29,314.43 \text{ MPa}}$$

$$A = 0.57 \text{ mm}^2$$

$$A_p = A(SF) = 0.57 \text{ mm}^2(4) = 2.28 \text{ mm}^2$$

$$A \text{ of Circle} = \frac{\pi d^2}{4}$$

$$\text{Solve for } d = \sqrt{\frac{A(4)}{\pi}}$$

$$d = \sqrt{\frac{A_p(4)}{\pi}}$$

$$d = \sqrt{\frac{2.28 \text{ mm}^2(4)}{\pi}}$$

$$d = 1.70 \text{ mm} \rightarrow \text{plug into CAD}$$